

# **ENTERPRISE OPERATING SYSTEM COMPARISON**

## **Week 3 - Enterprise Server Deployment**

ITEP 414 - System Administration and Maintenance

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## Windows Server vs Ubuntu Server vs Rocky Linux

| Category               | Windows Server                                                                                                          | Ubuntu Server                                                                              | Rocky Linux                                                                                                      |
|------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Licensing              | Commercial Microsoft licensing; evaluation edition is time-limited.                                                     | Open-source; free to use, with optional paid Ubuntu Pro/support.                           | Open-source community Enterprise Linux; no license fee for the OS.                                               |
| User Interface         | Server Core or Desktop Experience GUI.                                                                                  | Primarily command-line on Server; optional GUI packages.                                   | Primarily command-line for servers; desktop environments are available.                                          |
| Package Management     | Windows Update, PowerShell, roles/features, package tools.                                                              | APT / dpkg with Ubuntu repositories.                                                       | DNF / RPM with Enterprise Linux repositories.                                                                    |
| Security               | Active Directory ecosystem, Defender/security baselines, Windows Firewall, BitLocker and enterprise management options. | AppArmor, UFW/nftables, SSH, unattended/security updates, strong Linux permissions.        | SELinux, firewalld/nftables, SSH, Enterprise Linux security model.                                               |
| Performance            | Strong for Microsoft workloads and Windows-native enterprise applications; GUI edition has additional overhead.         | Efficient server footprint; common for web, database, container, and cloud workloads.      | Stable enterprise-oriented Linux platform suited to long-running server workloads.                               |
| Typical Enterprise Use | Active Directory, Windows file services, IIS, .NET, Microsoft-centric infrastructure.                                   | Web/database servers, cloud workloads, DevOps, containers, general Linux infrastructure.   | RHEL-compatible enterprise servers, hosting, databases, infrastructure requiring Enterprise Linux compatibility. |
| Advantages             | Deep Microsoft integration, mature GUI/admin ecosystem, AD and Group Policy.                                            | No OS license fee, large community, strong cloud/container ecosystem, LTS releases.        | Enterprise Linux compatibility, long-term stability focus, community-driven and open-source.                     |
| Disadvantages          | Licensing cost/complexity; Windows-specific administration overhead.                                                    | Some proprietary enterprise applications may not support Linux; CLI skills often required. | Smaller desktop/general-user ecosystem; some commercial vendors certify RHEL before Rocky Linux.                 |

## Discussion

Windows Server, Ubuntu Server, and Rocky Linux can all support enterprise infrastructure, but they are optimized for somewhat different environments. Windows Server is strongest when an organization depends heavily on Microsoft technologies such as Active Directory, Group Policy, Windows-based applications, and a GUI-oriented administration ecosystem. Its commercial licensing is a significant planning factor, but the platform provides mature tools for Microsoft-centered organizations.

Ubuntu Server is a general-purpose Linux server platform with a large software ecosystem and Long Term Support releases. It is widely suited to web services, databases, development environments, containers, cloud

workloads, automation, and remote administration. Because the operating system is open-source, organizations can deploy it without a per-server operating-system license, while optional commercial support is available. Its command-line-first administration model is efficient but requires Linux skills.

Rocky Linux is also open-source, but its identity is centered on community Enterprise Linux. It is designed for organizations that want an Enterprise Linux environment and compatibility with the broader RHEL ecosystem. It uses RPM/DNF package management and technologies such as SELinux that are common in Enterprise Linux administration.

The best choice therefore depends on workload and organizational requirements rather than one operating system being universally superior. A Microsoft-focused company may select Windows Server for directory and application integration. A cloud-native development team may prefer Ubuntu Server. An organization standardizing around Enterprise Linux practices may prefer Rocky Linux. System Administrators should understand all three because mixed operating-system environments are common, and licensing, application compatibility, staff expertise, security requirements, support expectations, and automation tooling all influence the final decision.

## References

Microsoft Windows Server documentation - <https://learn.microsoft.com/windows-server/>

Ubuntu Server documentation - <https://help.ubuntu.com/>

Rocky Linux documentation - <https://docs.rockylinux.org/>